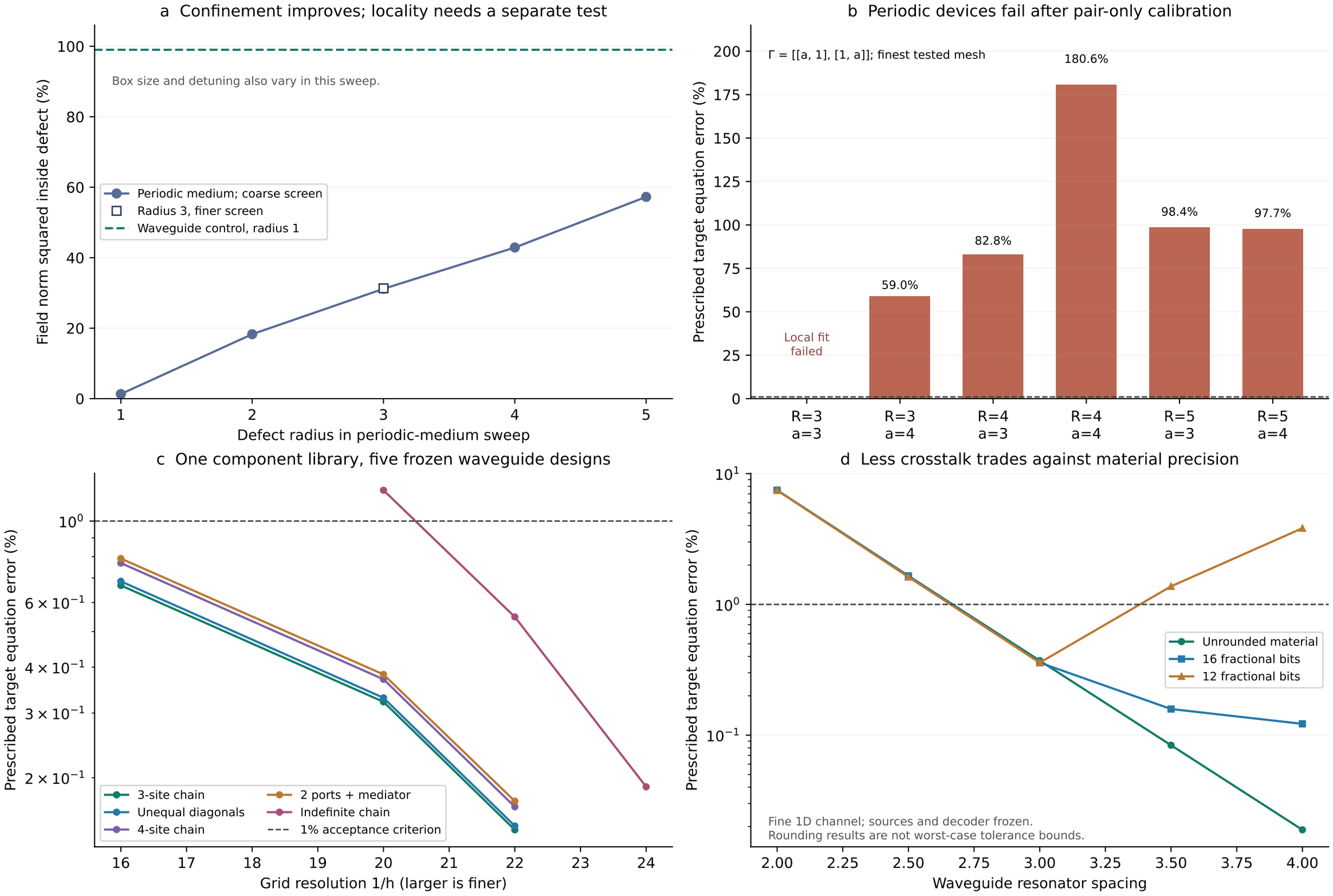
a Confinement improves; locality needs a separate test
Box size and detuning also vary in this sweep.
Periodic medium; coarse screen
Radius 3, finer screen
Waveguide control, radius 1
Field norm squared inside defect (%)
Defect radius in periodic-medium sweep
b Periodic devices fail after pair-only calibration
Γ = [[a, 1], [1, a]]; finest tested mesh
Prescribed target equation error (%)
Local fit failed
59.0%
82.8%
180.6%
98.4%
97.7%
R=3 a=3
R=3 a=4
R=4 a=3
R=4 a=4
R=5 a=3
R=5 a=4
c One component library, five frozen waveguide designs
Prescribed target equation error (%)
3-site chain
Unequal diagonals
4-site chain
2 ports + mediator
Indefinite chain
1% acceptance criterion
Grid resolution 1/h (larger is finer)
d Less crosstalk trades against material precision
Prescribed target equation error (%)
Unrounded material
16 fractional bits
12 fractional bits
Fine 1D channel; sources and decoder frozen.
Rounding results are not worst-case tolerance bounds.
Waveguide resonator spacing